

Prompt Engineering

Query Transformation

Haresh Vadali



Introduction to Query Transformation

Query Rewrite

HyDE

Step Back Prompting

Comparative Analysis

Advantages and Limitations

Key Takeaways

What is Query Transformation?

Query Transformation is the process of modifying or enriching a user's query before retrieval or generation to improve the relevance and quality of responses.

- Acts as a **pre-processing step** in Retrieval-Augmented Generation (RAG) pipelines.
- Helps bridge the gap between **user intent** and **retrievable content**.
- Reduces ambiguity and enhances the semantic understanding of queries.

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🎯 Goal:

Improve retrieval accuracy and contextual relevance by transforming the query into a more effective form for downstream tasks.

Applications

- Enhancing retrieval relevance in RAG pipelines
- Reducing hallucinations in LLMs
- Improving domain-specific QA systems

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Impact

Combining these transformations enables robust, contextually aligned, and semantically rich retrieval.

Query Rewriting

Reformulates ambiguous or incomplete queries into clearer, semantically equivalent versions.

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Step-Back Prompting

Reformulates the query into a higher-level or conceptual version to guide retrieval toward more general, insightful results.

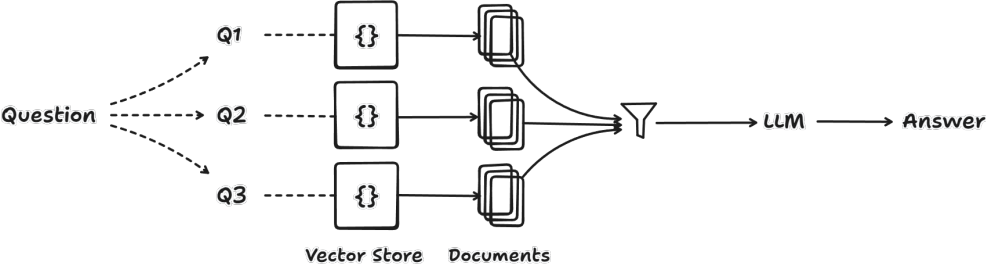
What is Query Rewriting?

A technique that transforms user queries to improve retrieval quality in RAG systems.

- **Reformulation:** Rephrasing queries for clarity
- **Expansion:** Adding relevant terms/synonyms
- **Simplification:** Removing ambiguity
- **Translation:** Converting to retrieval-friendly format

Core Principle

Bridge the gap between how users ask questions and how information is stored in the knowledge base.



Original Query

"Who was the first person to climb Mt. Everest?"

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Rewritten Queries

1. "Who successfully completed the first ascent of Mount Everest?"
2. "Which mountaineer first reached Everest's summit?"
3. "When was Mount Everest first climbed and by whom?"
4. "Details about the first successful Everest expedition"

What is HyDE?

A novel retrieval technique that generates hypothetical answer documents first, then uses them for retrieval.

- ❓ Take the user's **question**
- 💡 Generate a **hypothetical answer** (may be incorrect)
- 📄 Embed the **hypothetical document**
- 🔍 Search using this **embedding**
- ✅ Retrieve **actual documents**

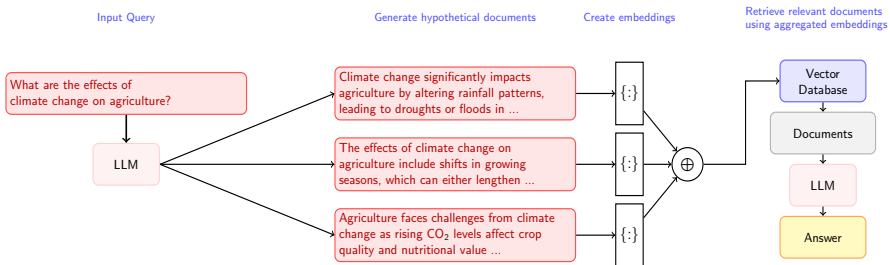
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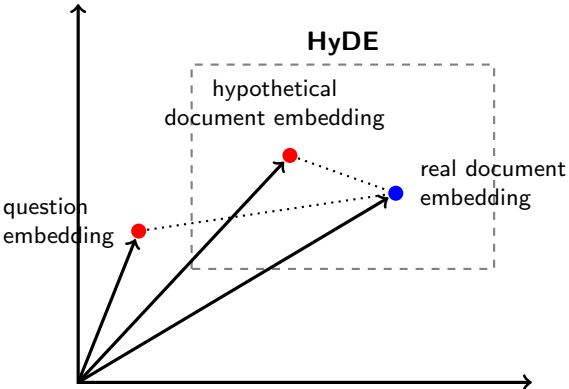
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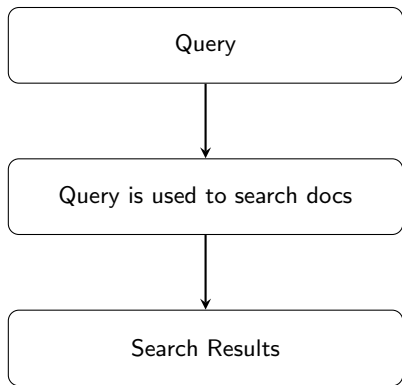
Key Insight

Documents are more similar to other documents than to questions. HyDE bridges this gap by searching with document-like text instead of questions.

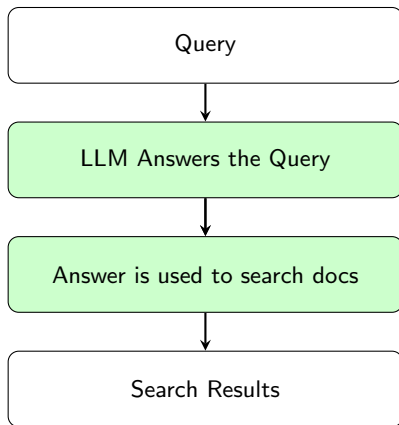




Standard RAG



HyDE RAG



What is Step-Back Prompting?

A technique that improves RAG systems by:

- **Abstracting** specific questions into general ones
- **Retrieving** broader contextual knowledge
- **Combining** principles with specifics
- **Generating** more informed answers

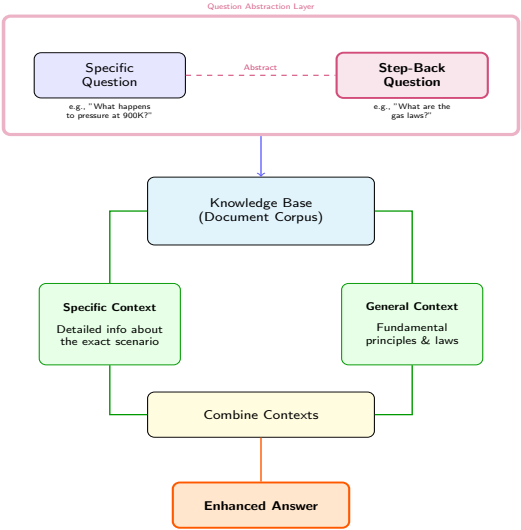
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Core Principle

Instead of diving into details immediately, *step back* to understand fundamental principles first — then apply them to specific cases.



Feature Comparison

- **Query Rewrite:** Best for semantic expansion
- **HyDE:** Excellent for complex information needs
- **Step Back:** Superior for conceptual understanding

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Usage Guidelines

- **Query Rewrite:** When query is ambiguous or poorly formulated
- **HyDE:** For detailed, specific information retrieval
- **Step Back:** When broader context is needed

Benefits

- Improved retrieval accuracy
- Better handling of complex queries
- Reduced hallucinations
- Enhanced context understanding
- More relevant results

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Challenges

- Computational overhead
- Potential loss of specificity
- Dependency on LLM quality
- Resource intensive
- May introduce latency

Key Takeaways

- Query transformation is crucial for RAG systems
- Different techniques suit different use cases
- Combination of approaches often yields best results
- Continuous evolution and improvement needed

Thank you for your attention!

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